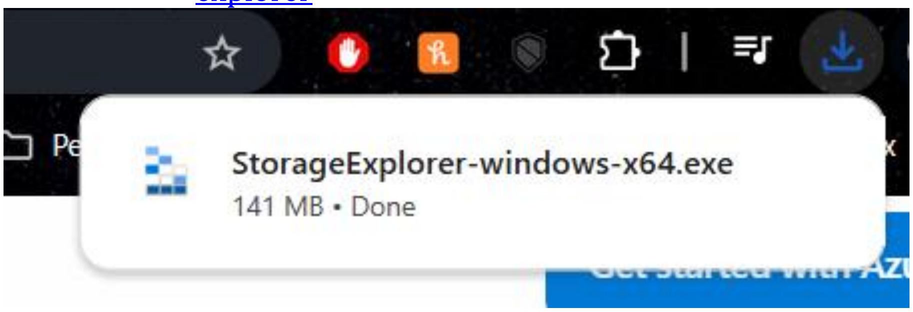
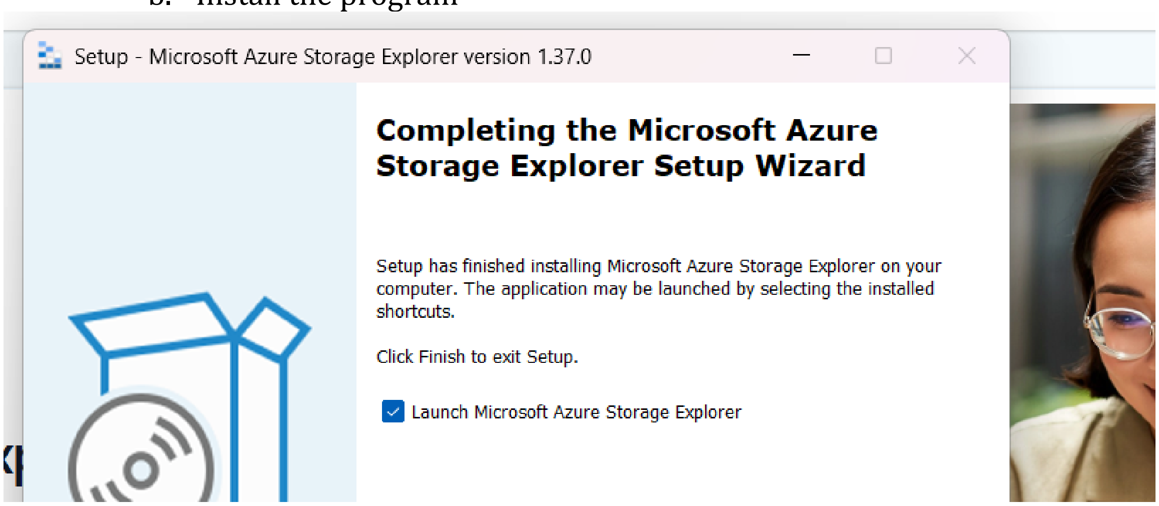
Juan Pablo de Legarreta
NETC-240
Lab 3 Part 1 - 1-29-2025

1. Download the Azure File Explorer
    a. https://azure.microsoft.com/en-us/products/storage/storage-explorer



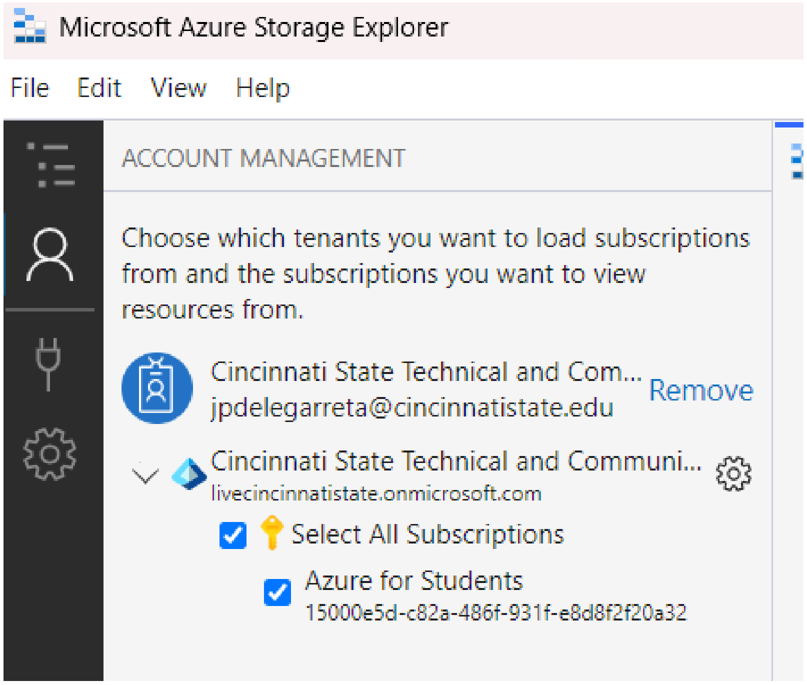
Successfully downloaded Azure Storage Explorer.

    b. Install the program



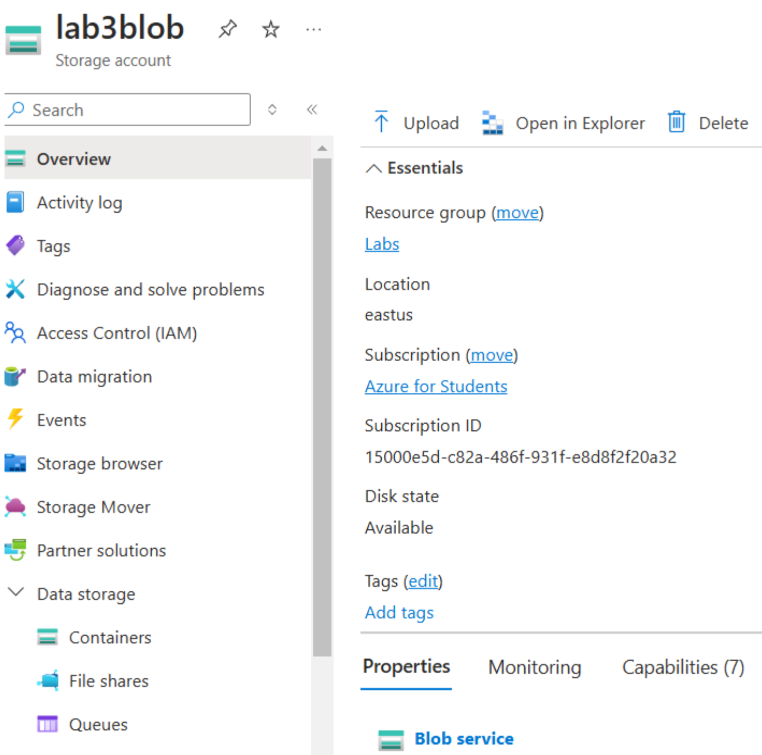
Successfully installed Azure Storage Explorer.

    c. Using the program, sign in to your Azure account



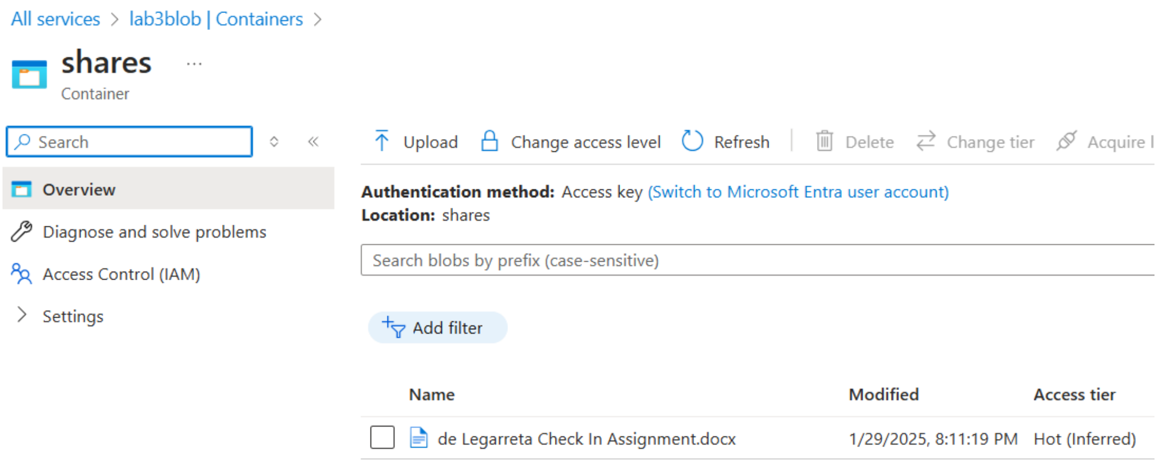
Successfully logged in with my Azure account tied to my Cincinnati State email address.

2. Create an Azure BLOB

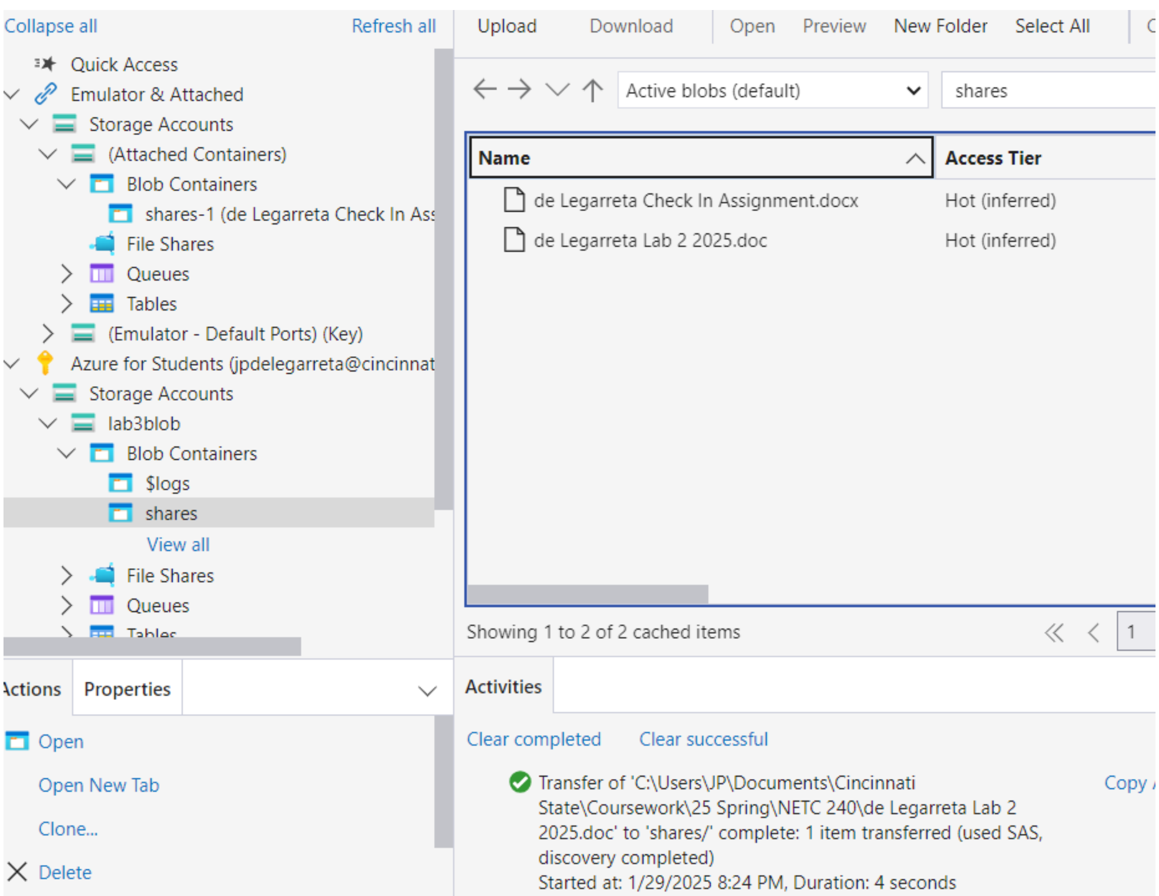


Successfully created a blob called lab3blob.

3. Use Azure Storage Explorer to upload a file to your Blob



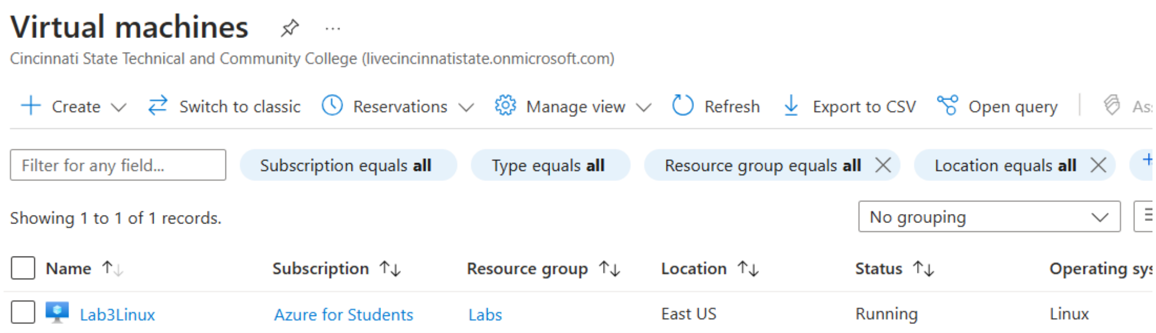
I uploaded this first file on the browser interface.



I uploaded the second file, de Legarreta Lab 2 2025, using the Azure Storage Explorer.

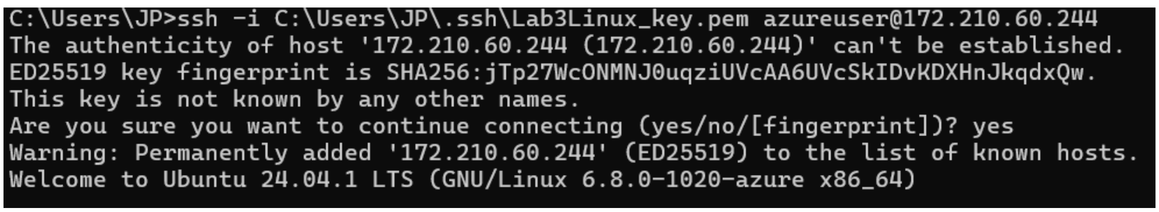
Part 2

1. Create a Ubuntu 24.04 LTS VM.

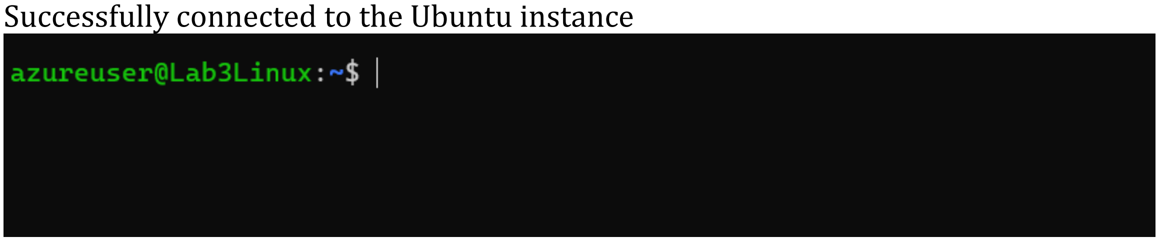


Successfully created a Ubuntu 24.04 LTS VM.

2. Connect to the VM using SSH and .pem file

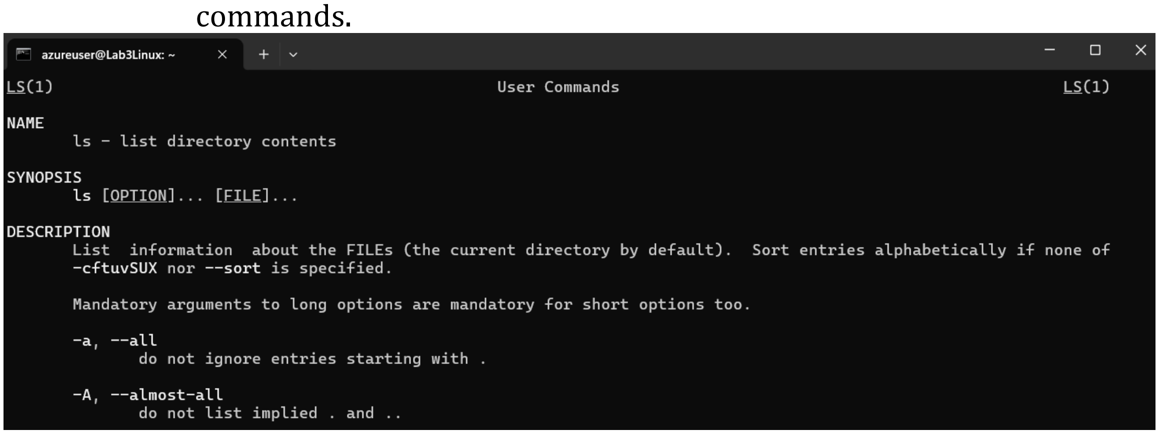


Successfully connected to the Ubuntu instance

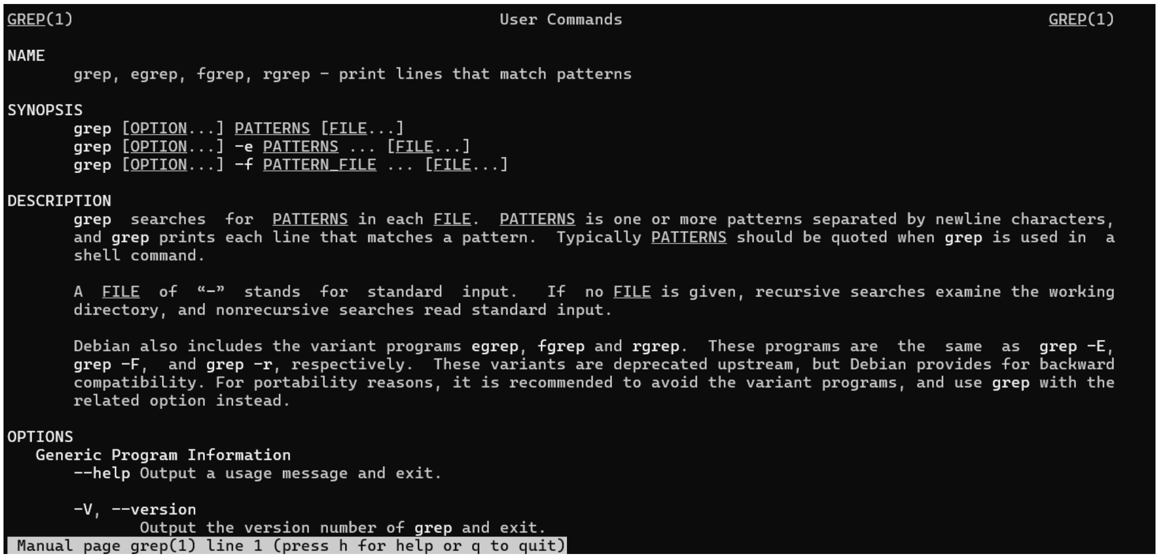


A screenshot of the command line with username

3. Use the following commands:
   a. To access Linux help, use the man xxx command. Such as man ls That will give you help on the "ls" command. Use this for 3 different commands.

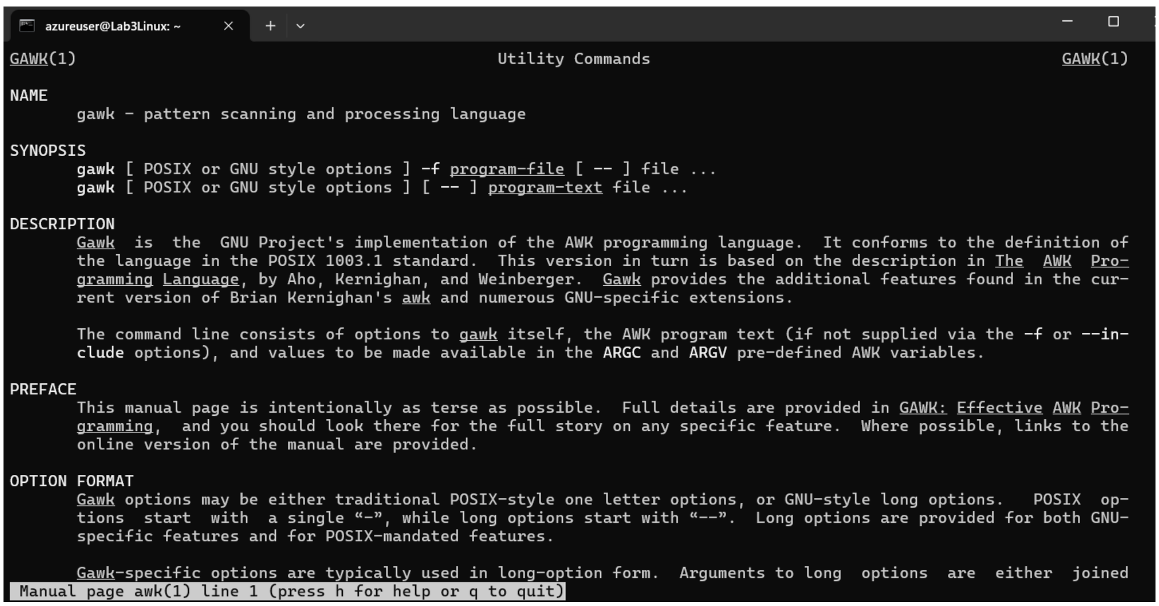


A screenshot of man ls



```
GREP(1)                         User Commands                         GREP(1)

NAME
       grep, egrep, fgrep, rgrep - print lines that match patterns

SYNOPSIS
       grep [OPTION...] PATTERNS [FILE...]
       grep [OPTION...] -e PATTERNS ... [FILE...]
       grep [OPTION...] -f PATTERN_FILE ... [FILE...]

DESCRIPTION
       grep  searches  for  PATTERNS in each FILE.  PATTERNS is one or more patterns separated by newline characters,
       and grep prints each line that matches a pattern.  Typically PATTERNS should be quoted when grep is used in  a
       shell command.

       A  FILE  of  “-”  stands  for  standard  input.   If  no FILE is given, recursive searches examine the working
       directory, and nonrecursive searches read standard input.

       Debian also includes the variant programs egrep, fgrep and rgrep.  These programs are  the  same  as  grep -E,
       grep -F,  and grep -r, respectively.  These variants are deprecated upstream, but Debian provides for backward
       compatibility. For portability reasons, it is recommended to avoid the variant programs, and use grep with the
       related option instead.

OPTIONS
   Generic Program Information
       --help Output a usage message and exit.

       -V, --version
              Output the version number of grep and exit.
 Manual page grep(1) line 1 (press h for help or q to quit)
```

A screenshot of man grep



```
azureuser@Lab3Linux: ~

GAWK(1)                         Utility Commands                         GAWK(1)

NAME
       gawk - pattern scanning and processing language

SYNOPSIS
       gawk [ POSIX or GNU style options ] -f program-file [ -- ] file ...
       gawk [ POSIX or GNU style options ] [ -- ] program-text file ...

DESCRIPTION
       Gawk  is  the  GNU Project's implementation of the AWK programming language.  It conforms to the definition of
       the language in the POSIX 1003.1 standard.  This version in turn is based on the description in The  AWK  Pro-
       gramming Language, by Aho, Kernighan, and Weinberger.  Gawk provides the additional features found in the cur-
       rent version of Brian Kernighan's awk and numerous GNU-specific extensions.

       The command line consists of options to gawk itself, the AWK program text (if not supplied via the -f or --in-
       clude options), and values to be made available in the ARGC and ARGV pre-defined AWK variables.

PREFACE
       This manual page is intentionally as terse as possible.  Full details are provided in GAWK: Effective AWK Pro-
       gramming,  and you should look there for the full story on any specific feature.  Where possible, links to the
       online version of the manual are provided.

OPTION FORMAT
       Gawk options may be either traditional POSIX-style one letter options, or GNU-style long options.   POSIX  op-
       tions  start  with  a single “-”, while long options start with “--”.  Long options are provided for both GNU-
       specific features and for POSIX-mandated features.

       Gawk-specific options are typically used in long-option form.  Arguments to long  options  are  either  joined
 Manual page awk(1) line 1 (press h for help or q to quit)
```

A screenshot of man gawk

4. Modify the VM to allow users to login with a username and password
   a. Modify the /etc/ssh/sshd_config file using Nano
   b. Set the PasswordAuthentication yes (default is no)
   c. Set the ChallengeResponseAuthentication yes (default is no)
   d. Save the file

```
# To disable tunneled clear text passwords, change to no here!
PasswordAuthentication yes
PermitEmptyPasswords no

# Change to yes to enable challenge-response passwords (beware issues with
# some PAM modules and threads)
KbdInteractiveAuthentication yes
```

A screenshot of the relevant changes to the sshd_config file.

e. Restart the SSH daemon. sudo systemctl restart ssh

```
azureuser@Lab3Linux:/etc/ssh$ sudo systemctl restart ssh
azureuser@Lab3Linux:/etc/ssh$ |
```

A screenshot of the restarting of the ssh daemon

5. Create a new user with your first name.

```
azureuser@Lab3Linux:/$ sudo adduser juan_pablo
info: Adding user `juan_pablo' ...
info: Selecting UID/GID from range 1000 to 59999 ...
info: Adding new group `juan_pablo' (1002) ...
info: Adding new user `juan_pablo' (1002) with group `juan_pablo (1002)' ...
info: Creating home directory `/home/juan_pablo' ...
info: Copying files from `/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for juan_pablo
Enter the new value, or press ENTER for the default
        Full Name []:
        Room Number []:
        Work Phone []:
        Home Phone []:
        Other []:
Is the information correct? [Y/n] Y
info: Adding new user `juan_pablo' to supplemental / extra groups `users' ..
.
info: Adding user `juan_pablo' to group `users' ...
```

A successful addition of user juan_pablo.

6. Create another new user, using your last name

```
azureuser@Lab3Linux:/etc/ssh$ sudo adduser de_legarreta
info: Adding user `de_legarreta' ...
info: Selecting UID/GID from range 1000 to 59999 ...
info: Adding new group `de_legarreta' (1001) ...
info: Adding new user `de_legarreta' (1001) with group `de_legarreta (1001)'
 ...
info: Creating home directory `/home/de_legarreta' ...
info: Copying files from `/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for de_legarreta
Enter the new value, or press ENTER for the default
        Full Name []:
        Room Number []:
        Work Phone []:
        Home Phone []:
        Other []:
Is the information correct? [Y/n] Y
info: Adding new user `de_legarreta' to supplemental / extra groups `users'
...
info: Adding user `de_legarreta' to group `users' ...
```
Successful addition of the de_legarreta username.

7. Open two new SSH connections, in addition to #2 above. Login using your first name and then use your last name for the second login. You should now have 3 open SSH connections.

```
azureuser@Lab3L    juan_pablo@Lab:    de_legarreta@Lal

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Install the latest PowerShell for new features and improvements
.ms/PSWindows

PS C:\Users\JP> ssh juan_pablo@172.210.60.244
(juan_pablo@172.210.60.244) Password:
Welcome to Ubuntu 24.04.1 LTS (GNU/Linux 6.8.0-1020-azure x86_6

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
```

```
azureuser@Lab3L    juan_pablo@Lab3    de_legarreta@La

de_legarreta@Lab3Linux:~$
```

The first image is a screenshot of all three ssh terminals open, one for azureusers, one for juan_pablo, one for de_legarreta. The second screenshot is the cleared screen for the de_legarreta user.

8. Promote your first name user to give them root privileges. Show how you verify that the user now has root privileges.

```
azureuser@Lab3Linux:/$ sudo usermod -aG sudo juan_pablo
azureuser@Lab3Linux:/$
```

This command adds the user juan_pablo to the user group 'sudo', giving it root privileges.

```
juan_pablo@Lab3Linux:~$ sudo -l -U juan_pablo
Matching Defaults entries for juan_pablo on Lab3Linux:
    env_reset, mail_badpass,
    secure_path=/usr/local/sbin\:/usr/local/bin\:/usr/sbin\:/usr/bin\:/sbin\
:/bin\:/snap/bin,
    use_pty

User juan_pablo may run the following commands on Lab3Linux:
    (ALL : ALL) ALL
```

This command shows the sudo information about the user juan_pablo, which displays that this user can use all commands, meaning it has sudo privileges.

9. The "cat" in Linux is similar to "type" in Windows. It types the contents of a file to the screen (standard output)
   a. Try cat groups   This shows info on all of the users

```
azureuser@Lab3Linux:/etc$ cat group
root:x:0:
daemon:x:1:
bin:x:2:
sys:x:3:
adm:x:4:syslog,azureuser
tty:x:5:
disk:x:6:
lp:x:7:
mail:x:8:
news:x:9:
uucp:x:10:
man:x:12:
proxy:x:13:
kmem:x:15:
dialout:x:20:
fax:x:21:
voice:x:22:
cdrom:x:24:azureuser
floppy:x:25:
tape:x:26:
sudo:x:27:azureuser,juan_pablo
```

When we cat group in the /etc folder, we get important information about the users.

   b. Try grep a /etc/group   Explain what grep is doing

```
azureuser@Lab3Linux:/$ grep a /etc/group
daemon:x:1:
adm:x:4:syslog,azureuser
mail:x:8:
man:x:12:
dialout:x:20:
fax:x:21:
cdrom:x:24:azureuser
tape:x:26:
sudo:x:27:azureuser,juan_pablo
audio:x:29:
dip:x:30:azureuser
www-data:x:33:
backup:x:34:
operator:x:37:
shadow:x:42:
sasl:x:45:
staff:x:50:
games:x:60:
users:x:100:de_legarreta,juan_pablo
systemd-journal:x:999:
crontab:x:997:
messagebus:x:101:
lxd:x:105:azureuser
rdma:x:107:
landscape:x:109:
admin:x:110:
azureuser:x:1000:
de_legarreta:x:1001:
juan_pablo:x:1002:
```

Grep a /etc/group is retrieving all instances of the character ‘a’ within the document group in the etc folder, as shown in the screenshot above.

c. Now try grep root /etc/group What information about the user does it provide?

```
azureuser@Lab3Linux:/$ grep root /etc/group
root:x:0:
```

This screenshot shows the root user's information, including its group name (root), that it uses shadow passwords (x), and its group ID number (0).

10. Determine the IP address of the VM from the CLI. Show the command that you used.

```
azureuser@Lab3Linux:/$ dig +short myip.opendns.com @resolver1.opendns.com
172.210.60.244
azureuser@Lab3Linux:/$
```

The IP address is 172.210.60.244, confirmed by the original ssh command and the azure screenshots.

11. Use the command the following commands:
    a. Arp

```
azureuser@Lab3Linux:~$ arp -v
Address                  HWtype  HWaddress           Flags Mask            Iface
_gateway                 ether   12:34:56:78:9a:bc   C                     eth0
Entries: 1      Skipped: 0      Found: 1
```

A screenshot of the Arp command.

    b. Ping

```
azureuser@Lab3Linux:/$ ping 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
64 bytes from 8.8.8.8: icmp_seq=1 ttl=54 time=1.13 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=54 time=1.28 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=54 time=1.09 ms
64 bytes from 8.8.8.8: icmp_seq=4 ttl=54 time=1.16 ms
64 bytes from 8.8.8.8: icmp_seq=5 ttl=54 time=1.14 ms
64 bytes from 8.8.8.8: icmp_seq=6 ttl=54 time=1.08 ms
64 bytes from 8.8.8.8: icmp_seq=7 ttl=54 time=1.10 ms
64 bytes from 8.8.8.8: icmp_seq=8 ttl=54 time=1.24 ms
^C
--- 8.8.8.8 ping statistics ---
8 packets transmitted, 8 received, 0% packet loss, time 7009ms
rtt min/avg/max/mdev = 1.081/1.152/1.278/0.067 ms
```

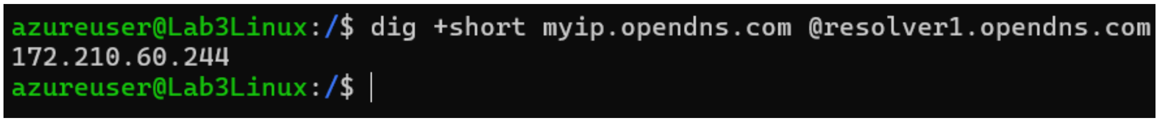
Pinged google at 8.8.8.8.

    c. Traceroute

```
azureuser@Lab3Linux:~$ traceroute 8.8.8.8
traceroute to 8.8.8.8 (8.8.8.8), 64 hops max
  1   *  *  *
  2   *  *  *
  3   *  *  *
  4   *  *  *
  5   *  *  *
```

This is showing the hops that packets take to get to 8.8.8.8.

    d. Dig

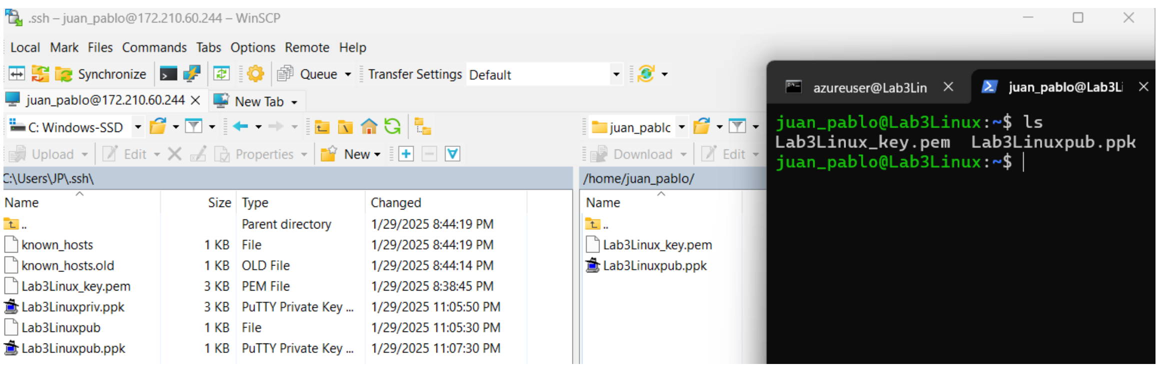


This particular use of dig gets the public ip address from myip.opendns.com

e. Hostname

Hostname displays the name of the computer

12. Use a SFTP client, such as WinSCP to connect to your VM.
    a. Upload a copy of your .pem file and public .ppk file to your home directory



Here is a screenshot of the WinSCP client with the two files moved over, reflected by the ls command on the home directory of the user juan_pablo.

Questions:
1. What is the command to create a new user?

adduser is the command to add a new user, though you might need to use sudo.

2. What are AWK, SED and GREP. Explain what each one does.

GREP, or Global Regular Expression Print, is a command that you can use to search and return text patterns in files. It can retrieve strings, numbers, etc. and print them.

SED, or Steam Editor, is similar to GREP but with more functionality – it can also substitute text, modify files, and add basic scripts.

AWK, or Aho, Weinberger & Kernighan, is the most powerful of the three, as it is a scripting language, comparable to Perl, used for pattern scanning and processing. AWK's built-in functions as well as providing the capability for users to write their own make it a very powerful tool.

Source: https://www.baeldung.com/linux/grep-sed-awk-differences

3. What is “sudo” and how and when is it used?

Sudo means ‘super user do’, and it’s Linux’s version of ‘run as administrator’ in windows. It allows for root privileges to be used with certain commands. Some commands will require sudo to be used, such as installing software. (sudo apt install)

4. How can you determine the IP address from the CLI?

The dig command can be used to determine the ip address from the CLI, namely by using:

dig +short myip.opendns.com @resolver1.opendns.com

5. How can you verify that DNS is functioning properly?

You can use the dig command on a website and it should tell you what server the dns resolved the URL to. So, for example, if you type:

dig google.com

it will show a SERVER with an IP address that should be the local DNS server.

6. What is a “daemon”? Explain what it is in Linux and what the parallel is in Windows OS

Dating back to the middle ages, a ‘daemon’ referred to a spirit. Inspired, no doubt, by this ancient meaning, a daemon in Linux refers to a program that runs in the background without direct user input. Usually, these functions are related to the operating system or handling network requests. In Linux, these processes usually end in ‘d’ such as syslogd or sshd (the syslog and ssh daemons, respectively).

In Windows, these processes are called Windows services.